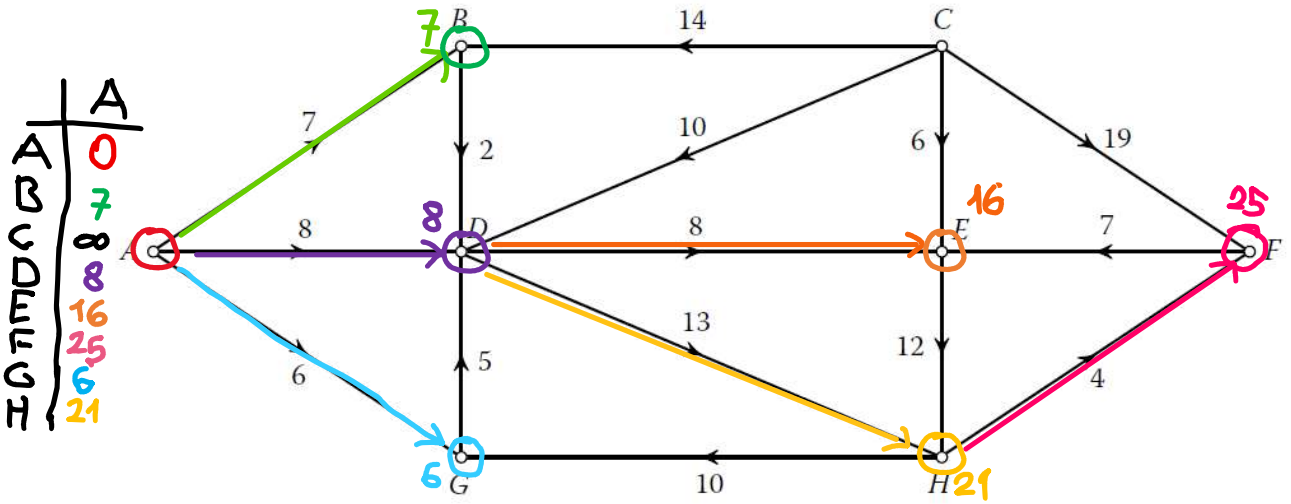
CSE 5004-Graph Theory and Algorithms

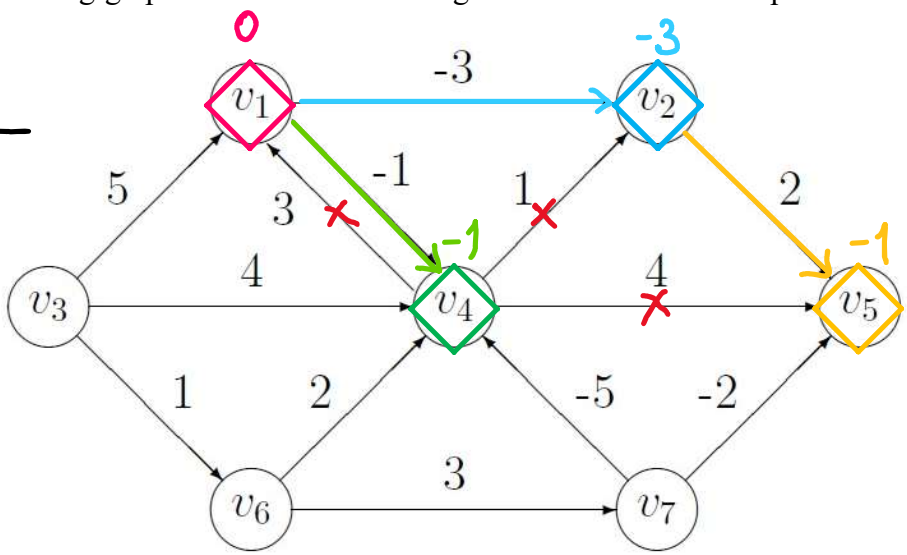
Exercise – 5

1. In the graph given below use Dijkstra's algorithm to find shortest paths from A to other nodes.



	A
A	0
B	7
C	∞
D	8
E	16
F	25
G	6
H	21

2. In the following graph use Bellman-Ford algorithm to find shortest paths from v_1 to other nodes.



	V_1
V_1	0
V_2	-3
V_3	∞
V_4	-1
V_5	-1
V_6	∞
V_7	∞

3. Test the Floyd-Warshall Algorithm on:
https://algorithms.discrete.ma.tum.de/graph-algorithms/spp-floyd-warshall/index_en.html

- Create a graph from "Create Graph" tab. Modify the default graph to the graph given in Q1.
  - To create a node, make a double-click in the drawing area.
  - To create an edge, first click on the output node and then click on the destination node.
  - The edge weight can be changed by double clicking on the edge.
  - Right-clicking deletes edges and nodes.

- Execute the algorithm as following:

  Ready – Run the Algorithm! → Let's go! → next

- Paste the final distance matrix.

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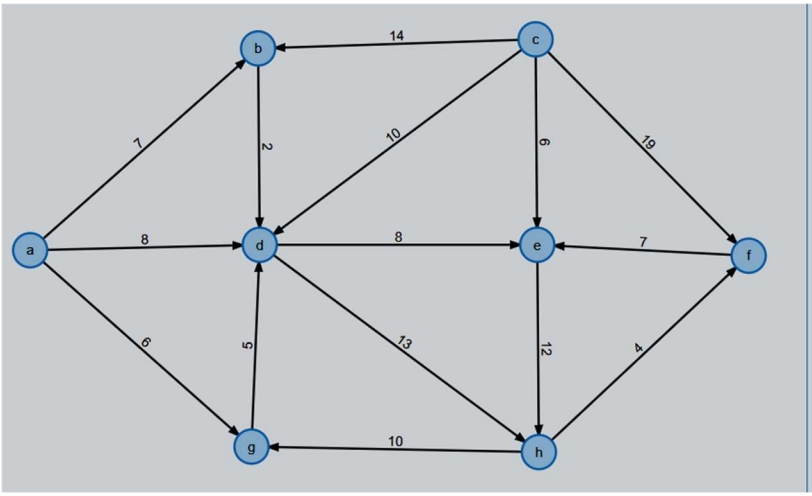

Current state of the algorithm

prev | next | fast forward

Description | **Pseudocode**

End of the algorithm.

All shortest (or cheapest, respectively) paths have been computed.

The final matrix looks as follows:

	a	b	c	d	e	f	g	h
a	0	7	∞	8	16	25	6	21
b	∞	0	∞	2	10	19	25	15
c	∞	14	0	10	6	19	28	18
d	∞	∞	∞	0	8	17	23	13
e	∞	∞	∞	27	0	16	22	12
f	∞	∞	∞	34	7	0	29	19
g	∞	∞	∞	5	13	22	0	18
h	∞	∞	∞	15	11	4	10	0